

- Q.23 What are the five different manufacturer's in PLC?
- Q.24 Write down any five applications of PLC in instrumentation.
- Q.25 What is RTU? Write down its application.
- Q.26 Explain the working of DCS with diagram.
- Q.27 What is counter instruction? Discuss "up counter" and "up-down counter" instruction.
- Q.28 What are the disadvantages of DCS?
- Q.29 What is real time clock function? How does it work?
- Q.30 What are the types of timer instruction? Explain retentive and non-retentive timer instruction.
- Q.31 Explain the importance of HMI with its applications.
- Q.32 What is watch dog timer? How does it work?
- Q.33 Write down any five advantages of DCS.
- Q.34 Define comparison instruction. Explain "less than" and "greater than" instructions.
- Q.35 What is SCADA? How does it work?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the basic operation of DCS. Write down any four differences between SCADA and DCS. What are advantages of PLC?
- Q.37 What is SCADA system? Explain the applications of it. How it is different from PLC?
- Q.38 What is the function of human machine interface? Write down the advantages of it. Draw and explain the scan cycle in PLC.

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6th Sem / IC, EI

Subject:- PLC, DCS and SCADA/ Micro-contr & PLC Based Instr.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is not the building blocks of PLC?
- a) Input module b) Power supply
c) Processor d) None of the above
- Q.2 What type of programming language is typically used to program a PLC?
- a) C++ b) JAVA
c) ladder logic d) Python
- Q.3 What type of power supply is typically used for a PLC?
- a) AC b) Both AC and DC
c) DC d) Solar
- Q.4 HMI stands for
- a) Human machine interface
b) Human machine industry
c) Home machine interface
d) Human move interface

- Q.5 What is the difference between a PLC and a DCS?
- PLCs are used to control machines while DCSs are used to control entire plants
 - PLCs are used to control machines while DCSs are used to control networks
 - PLCs are used to control machines while DCSs are used to control buildings
 - PLCs are used to control machines while DCSs are used to control vehicles.
- Q.6 SUB instruction is
- An arithmetic instruction
 - A Timer instruction
 - A counter instruction
 - None of the above
- Q.7 What is the function of a PLC's high-speed counter?
- To count the number of pulses from a sensor
 - To measure the speed of a motor
 - To monitor the frequency of a signal
 - All of the above
- Q.8 DDC stands for
- Direct digital control
 - Distributed direct control
 - Distributed digital control
 - Digital data control
- Q.9 Which of the following timer only turns its output ON after a preset time DELAY
- On DELAY timer
 - Off DELAY timer
 - Retentive timer
 - None-retentive timer

- Q.10 _____ is used to create, modify or troubleshoot a program in PLC is
- ROM
 - Desktop
 - Power supply
 - Flash memory

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define DCS.
- Q.12 Expand RTU.
- Q.13 Functional block diagram is a programming language in PLC. (True/False)
- Q.14 The process of _____, running the control program, and updating outputs is known as scan cycle.
- Q.15 Expand RTC.
- Q.16 Define real time clock in PLC.
- Q.17 _____ system that uses software and hardware to monitor and control industrial processes and equipment.
- Q.18 _____ is a type of counter that increases its value by one each time when it receives an input pulse, until it reaches as preset limit.
- Q.19 Write any one application of DCS.
- Q.20 A processor is the part of block diagram of PLC. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the working of PLC using scan cycle.
- Q.22 Write any five differences between DCS and SCADA.